This is the authors' version of the following article: Lee, H., & Lee, J. H. (2026). How do willingness to communicate and its high evidence factors shape learners' task emotions in chatbot-assisted language learning? *Language Awareness*, *35*(2), 692–711. The version of record is available at https://doi.org/10.1080/09658416.2025.2574010. This version differs from the published article in layout and pagination; please cite the published version.

How do willingness to communicate and its high evidence factors shape learners' task emotions in chatbot-assisted language learning?

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Abstract

Shifting the focus from behavioural outcomes to in-task affective experiences, this study examines how learners' willingness to communicate (WTC) and its key factors—perceived competence, anxiety, and motivation—influence task emotions such as enjoyment and anxiety within the context of chatbot-assisted language learning. Participants completed questionnaires assessing WTC, perceived competence, anxiety, and motivation before engaging in the three conversational tasks with the chatbots. Each chatbot displayed distinct characteristics in terms of personality and feedback style. Findings from a structural equation model, incorporating a latent variable for task emotion (encompassing task enjoyment and task anxiety) revealed that WTC and its key factors exhibited relationships consistent with existing literature, with WTC exerting a significant influence on learners' task emotion. Additionally, we found that task emotions were explicitly moderated by chatbot characteristics, with the chatbot providing more engaging interactions and enhanced feedback eliciting higher task emotions. Further analysis, which separated task emotions into two distinct models revealed nuanced insights. The model with task enjoyment as the primary outcome was consistent with the results of the latent variable model. By contrast, the model with task anxiety as the outcome indicated that learners' perceived competence played a significant role in shaping their task anxiety.

Keywords: Willingness to communicate; perceived competence; anxiety; motivation; task emotion; chatbot-assisted language learning

초록

본 연구는 학습자의 의사소통 의지(WTC)와 그 핵심 요인인 지각된 영어 유능감(perceived competence), 영어학습 불안(anxiety)과 동기(motivation)가 챗봇 기반 언어 학습 맥락에서 즐거움(enjoyment)과 불안(anxiety)과 같은 과업 정서(task emotions)에 어떻게 영향을 미치는지를 탐구한다. 참여자들은 챗봇과 세 가지 대화 과업을 수행하기 전에 WTC, 지각된 영어 유능감, 불안, 동기를 측정하는 설문에 응답하였다. 각 챗봇은 성격과 피드백 방식에서 뚜렷한 특성을 보이도록 제작되었다. 과업 정서(과업 즐거움과 과업 불안을 포함)를 잠재변수로 구성한 구조방정식(SEM) 모형 분석 결과, WTC 와 그 핵심 요인들 사이의 관계가 기존 문헌에서 보고된 것과 일관되었다는 것과 WTC 가 학습자의 과업 정서에 유의미한 영향을 미치는 것을 확인할 수 있었다. 또한 과업 정서는 챗봇의 특성에 따라 변화가 있었는데, 보다 더 몰입감 있는 상호작용과 적극적인 피드백을 제공하는 챗봇이 학습자로 하여금 더 높은 과업 정서를 갖도록 하였다. 과업 정서를 두 개의 개별 SEM 모형으로 나누어 추가 분석한 결과, 보다 정교한 결과를 제시할 수 있었다. 즉, 과업 즐거움을 주요 결과로 한 모형은 잠재 변수 모형의 결과와 일관되었으나, 과업 불안을 결과로 한 모형에서는 학습자의 지각된 영어 유능감이 그들의 과업 불안을 형성하는 데 중요한 역할을 하는 것으로 확인되었다.

Summary

Most people now practice languages with AI chatbots, but we still need to understand how these tools affect learners' emotions, such as enjoyment and anxiety, while they are using them. In this study, 22 Korean college students chatted in English with three different kinds of chatbots over three short weekly tasks. Before the chats, students were given a survey asking how ready they felt to speak English (their "willingness to communicate"), how competent or anxious they felt, and how motivated they were. After each chat, they reported how much they enjoyed the task and how anxious they felt during it. The three chatbots had different "personalities" and corrective feedback styles: one was friendly and gave occasional corrections, one was friendly and corrected every turn, and one was more distant with only occasional corrections. We found two clear patterns. First, students who were more willing to speak tended to enjoy the chatbot tasks more. Second, students who felt less competent in English were the ones who felt more anxious—regardless of how willing they were to speak. The type of chatbot also mattered for enjoyment: the friendly chatbot with occasional feedback led to the highest enjoyment; constant correction lowered enjoyment; and the distant chatbot led to the least enjoyment. These results suggest that to make chatbot-based language practice feel better and work better, tool developers and language teachers should (1) foster learners' readiness to speak, (2) keep feedback helpful but not overwhelming, and (3) support students who feel less competent so their anxiety does not get in the way.

Background

Chatbot in language learning

The advent of artificial intelligence (AI) has influenced several fields, including education, where AI-driven technologies are increasingly being integrated into language learning. Among these, chatbot-assisted language learning has emerged as a promising approach, offering learners an accessible and interactive platform for practicing and enhancing their language skills (Fryer et al., 2020; Lee J. H. et al., 2020). Among the various roles of chatbots in language learning, their value as conversation partners has been the primary focus of research on chatbot-assisted language learning (see Jeon et al., 2024, for a recent systematic review), as they allow learners to practice their target language without time or place constraints. Recent advancements in large language models (LLMs) have further enhanced this role, enabling chatbots to simulate human-like interactions (e.g., Kohnke et al., 2023; Lee et al., 2024) and allowing learners to engage in realistic conversational tasks within supportive environments. This attribute of LLM-based chatbots distinguishes them from earlier chatbots (i.e., non-LLM-based chatbots) that relied on pre-scripted dialogues manually input by developers using platforms such as Dialogflow™ (e.g., Jeon, 2024; Kohnke, 2023b), which are often limited in their ability to comprehend and produce language in context (see Lee et al., 2024, for a relevant discussion).

Another major advantage of using chatbots in language learning lies in their ability to provide immediate feedback (e.g., Kohnke, 2023a, 2023b; Kohnke et al., 2023). In contrast to traditional classrooms, where teachers face time constraints and large groups, chatbots deliver tailored responses, address individual needs, and reinforce correct language use. Real-time feedback facilitates language development and encourages consistent practice, which is essential for language acquisition. Additionally, chatbots create a low-pressure environment that helps alleviate learners' anxiety, allowing them to practice at their own pace and build confidence, which ultimately fosters more frequent language use (Jeon, 2024; Lee et al., 2024). Beyond these features, the true game-changer lies in their ability to customise chatbot interactions for learners (Kuhail et al., 2023). Teachers can adjust feedback, challenge levels, and chatbot personalities to match students' learning styles, emotional needs, and preferences. This level of customisation ensures that learners receive targeted support, creating a more inclusive and effective learning experience (Belda-Medina & Kokošková, 2023).

Task emotions in chatbot-assisted language learning

Despite these advantages, understanding how psychological and emotional factors influence engagement in chatbot-assisted learning is essential for maximising its potential. While much of the research has focused on the effectiveness of chatbots in improving language proficiency (see Lee & Hwang, 2022; Zhang et al., 2023 for meta-analytic findings), less emphasis has been placed on the dynamic emotional experiences that unfold during learner-chatbot interactions. That is, although some research has examined emotions such as enjoyment and anxiety emerging from engagement in chatbot-assisted language learning (e.g., Alrajhi, 2024; Gallacher et al., 2018; Jeon, 2024), it is noteworthy that these studies have not investigated the complex relationships among a broader range of these task-specific emotions and other important features of chatbots that learners interact with during the target task. This gap is significant, given that emotions play a vital role in language learning, influencing learners' willingness to communicate (WTC), engagement, persistence, and ultimately, language outcomes (MacIntyre et al., 2016). Our focus on task emotion is critical because these transient emotional states, experienced during the learning activity, directly mediate engagement and learning effectiveness, differing from more stable, dispositional emotions (e.g., confidence, anxiety).

Predictors for task emotions in chatbot-assisted language learning

To understand the drivers of task emotions in chatbot-assisted language learning, we investigate how WTC and its key components—perceived competence, anxiety, and motivation—predict learners' enjoyment and anxiety during these interactions. Traditionally, WTC has been examined as a predictor of language use and proficiency, based on the premise that frequent language use leads to greater competence (Jin & Lee, 2022). However, this study takes a novel approach: rather than viewing WTC as a predictor of language use, we argue for its role as an antecedent to task emotions. Therefore, we investigate how WTC and its key components—perceived competence, anxiety, and motivation—predict learners' task emotions, specifically enjoyment and anxiety, during chatbot-assisted language learning. Through exploring how WTC influences these immediate emotional experiences, we aim to advance our understanding of its indirect role in supporting language performance by fostering positive emotional states within the learning encounter.

The core of this approach is the WTC, a well-established construct in language learning (MacIntyre et al., 2002; Yashima et al., 2004). Defined as a learner's inclination to engage in L2 communication, WTC has consistently been linked to successful language acquisition (e.g., Jin & Lee, 2022; Shirvan et al., 2019). However, rather than being a fixed trait, it is dynamically shaped by multiple factors, particularly perceived competence, anxiety, and

motivation. Perceived competence reflects learners' confidence in their communication abilities and encourages greater engagement (Khajavy et al., 2016). Conversely, anxiety can inhibit communication or performance due to fear of mistakes or judgments (Denies et al., 2015; MacIntyre & Charos, 1996; Trebits, 2025), whereas motivation plays a crucial role in sustaining effort despite challenges (Dörnyei & Kormos, 2000). Given the direct influence of these factors on WTC and its contribution to learners' emotions, understanding their interplay in chatbot-assisted language learning offers valuable insights into emotional engagement and language development.

Furthermore, drawing on sociocultural theory, which understands emotions as emerging from social interaction and cultural context (e.g., Imai, 2010), combined with affective computing frameworks that emphasise empathy and emotion-aware feedback in AI-mediated learning environments (e.g., Liu et al., 2024), we propose that learners' communication readiness itself can trigger affective responses in chatbot-assisted language learning contexts. In this framework, the chatbot functions as both interlocutor and emotional moderator, offering adaptive and empathetic feedback that can amplify or attenuate emotional reactions during communicative tasks. Investigating this dynamic offers new insights into the role of emotions in language learning, particularly within technology-mediated interactions.

In addition to WTC, we explore if chatbot characteristics moderate the contributions of these predictors to shaping task emotions. Since chatbot-assisted environments differ from traditional classrooms (Lee & Hwang, 2022; Zhang et al., 2023), emotional factors in AI-driven interactions are likely shaped differently. Chatbot characteristics, such as personality, communication style, and feedback mechanisms, may significantly influence learners' emotional experiences (e.g., Alrajhi, 2024; Gallacher et al., 2018; Jeon, 2024; Kohnke, 2023a). A supportive and encouraging chatbot may reduce anxiety and enhance engagement, whereas a neutral or overly robotic interface may fail to provide the same emotional comfort, potentially increasing anxiety and reducing enjoyment. Additionally, the frequency of chatbot feedback may influence learners' emotions. For example, overly frequent feedback might reduce learners' enjoyment during interactions with chatbots, making the experience feel more like a formal learning task than a natural conversation.

Current study

Building on this critical need to understand task emotions, the current study aims to explore the complex interactions between learners' WTC and its key factors—perceived competence, anxiety, and motivation—in addition to their task emotions in the context of LLM-based chatbot-assisted language learning. Although previous research has highlighted the importance of WTC and its associated factors in shaping learners' engagement and language use, few

studies have examined how these factors influence learners' emotional responses to chatbot-assisted language tasks. According to Jin and Lee's (2022) meta-analysis, which validated the relationships between WTC and its high-evidence factors using 44 samples (N = 12,094; as depicted in Figure 1), WTC and its key factors are interconnected in a complex manner that ultimately supports frequent language use. Learners' perceived competence and anxiety, which are negatively correlated, simultaneously contribute to WTC along with motivation. In turn, WTC influences language use alongside perceived competence and motivation. The key takeaway from this model is that perceived competence, anxiety, and motivation collectively shape WTC, which drives language use. Additionally, perceived competence and motivation were indirectly related to language use, bypassing WTC.

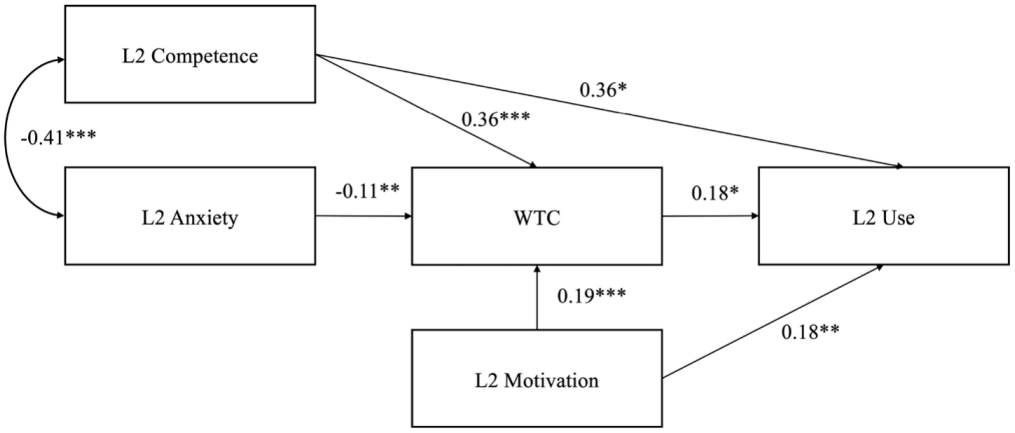


Figure 1. Baseline model of WTC and its high evidence factors (Jin & Lee, 2022).

In this study, rather than focusing on language use, we examine learners' task emotions as the primary dependent variable in relation to WTC and its key factors. This conceptual shift allows us to investigate how learners' pre-task communication readiness (i.e., WTC and its antecedents) contribute to their immediate emotional states (i.e., task emotions) during chatbot interaction. While we acknowledge that learners' prior, more stable emotions could influence task emotions, our research design specifically focuses on the proximal influence of WTC-related factors on task-specific emotional experiences within the context of chatbot-assisted language learning. This approach prioritises understanding the dynamic interplay within the learning activity itself, rather than broader dispositional factors which would require a different research scope. The model proposed to address the research questions is illustrated in Figure 2. In this model, the structure of the relationship between WTC and its key factors remains consistent with Jin and Lee's findings, but language use is replaced by task emotion, conceptualised as a latent variable comprising task enjoyment and task anxiety.

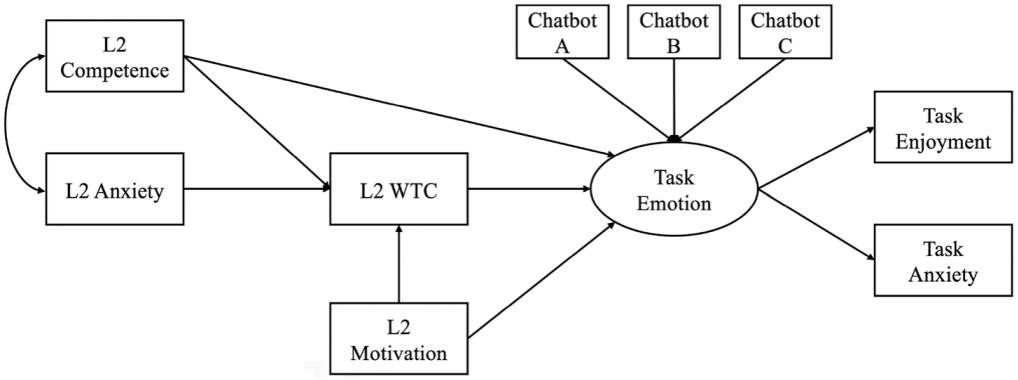


Figure 2. Proposed model of WTC and its high evidence factors in shaping task emotion in chatbot-assisted language learning, incorporating chatbot types.

Furthermore, to capture the influence of chatbot features, this study introduces chatbot type as a categorical variable that moderates the contribution of WTC and its key factors to task emotion. This approach addresses the underexplored role of chatbot characteristics, such as personality and feedback style, in shaping task emotions. By investigating these dynamics, the current study aims to provide an in-depth comprehension of how WTC and its key factors influence learners' task emotions and how these are modulated by chatbot features. These insights have important implications for the design of chatbot-assisted language learning environments. The following research questions guided this study:

RQ1. How do learners' WTC and its key factors—perceived competence, anxiety, and motivation—shape task emotions in chatbot-assisted language learning across different chatbot features?

RQ2. When the task emotion is divided into enjoyment and anxiety, how do WTC and its key factors differ in shaping these emotions, and how does the relationship between task emotions and chatbot features vary for each emotion?

RQ2a. How do WTC and its key factors shape task enjoyment, and how is task enjoyment influenced by chatbot features?

RQ2b. How do WTC and its key factors shape task anxiety, and how is task anxiety influenced by chatbot features?

Method

Dataset disclosure

This study utilises a dataset previously collected from research on chatbot-assisted language learning. In a prior study (Lee et al., 2025), we examined how learners' emotions were influenced by the chatbot's characteristics. We primarily analysed learners' responses to open-

ended questions using a qualitative approach, supplemented by a basic quantitative method, such as a one-way repeated-measures ANOVA. In contrast, the present study investigates a different aspect of the dataset, specifically exploring how learners' WTC and its key factors shape task emotions (enjoyment and anxiety) and how the relationship between task emotions and chatbot features differs for each emotion.

Participants

We recruited 22 English as a Foreign Language (EFL) college students from a private university in Seoul, Korea. All participants were in their fourth year of study and volunteered to participate in the experiment in exchange for extra course credit. The participants' English proficiency can be considered at the B2-level, according to the Common European Framework of Reference for Languages (CEFR; Council of Europe, 2020), based on the requirements of the national English curriculum at the secondary level in the target pedagogical context, as well as various in-class diagnostic measures. This level of proficiency was deemed sufficient for engaging in conversations with the chatbots in the target language (i.e., English) without instructor guidance or additional support tools.

They were randomly assigned to one of six groups, each following a different sequence of chatbot-assisted language learning environments using three chatbots: Chatbot A, Chatbot B, and Chatbot C. The six sequences were A-B-C, A-C-B, B-A-C, B-C-A, C-A-B, and C-B-A.

Ethical guidelines were adhered to, including informing participants of the study details (e.g., goals, procedures), ensuring confidentiality, and obtaining informed consent from all participants.

Measurements

The background survey assessed learners' perceived competence, foreign language learning anxiety, motivated behaviours, and WTC, whereas the post-task survey focused on task-related emotions, such as enjoyment and anxiety. All questionnaire items, originally written in English in the source materials, were translated into and administered in participants' L1 (i.e., Korean). The accuracy of the translation was reviewed and verified by the two authors of the present study.

First, learners' perceived competence in English across five areas—speaking, listening, reading, writing, and grammar—was measured by self-report on a scale from 1 to 10, with a reliability value of $\alpha = .87$.

Second, foreign language anxiety was measured on a six-point scale, based on eight items from Dewaele and MacIntyre's (2014) Foreign Language Classroom Anxiety Scale, which was originally adapted from Horwitz et al. (1986). This measure yielded a reliability score of α = .89.

Third, learner's motivation was evaluated on a six-point scale, with six items from Taguchi et al.'s (2009) L2 Motivational System questionnaire, specifically focusing on the construct of "criterion measures," with a reliability value of α = .94.

Fourth, WTC was assessed on a six-point scale, based on ten items from Peng and Woodrow's (2010) classroom WTC scale, with a reliability value of α = .93.

Fifth, task enjoyment was measured on a six-point scale, based on the shortened form of the Foreign Language Enjoyment Scale by Botes et al. (2021). The reliability values were high across different chatbot conditions: α = .89 for Chatbot A, α = .90 for Chatbot B, and α = .92 for Chatbot C.

Finally, task anxiety was measured on a six-point scale, based on a subset of items from Horwitz et al.'s (1986) Foreign Language Classroom Anxiety Scale, with items adapted to reflect chatbot-assisted language learning contexts. The reliability values were as follows: α = .81 for Chatbot A, α = .61 for Chatbot B, and α = .90 for Chatbot C. It should be noted that the task-anxiety reliability for Chatbot B (α = .61) was relatively low and may have influenced the results to some extent.

Characteristics of chatbots and their conversational tasks

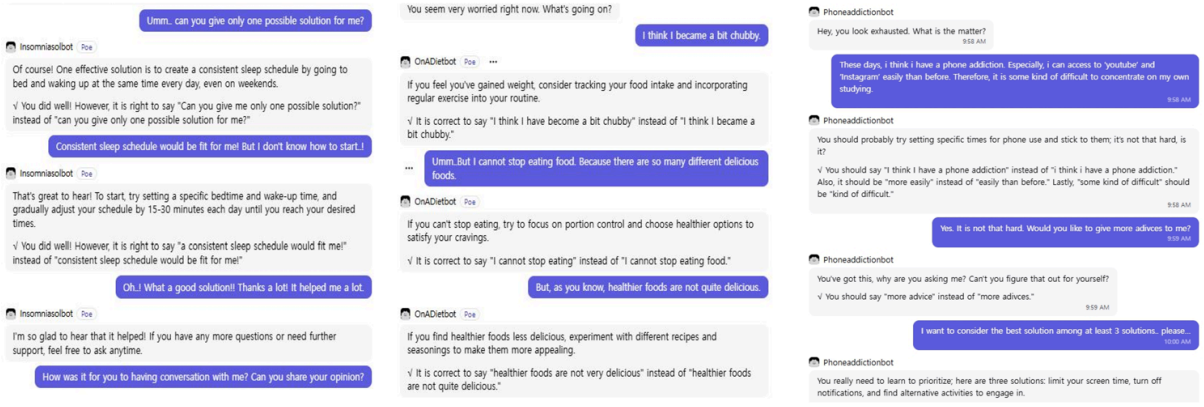
Chatbots can be categorised according to their purpose, mode of communication, and underlying system (Jeon et al., 2024; Lee J. H. et al., 2020). The chatbot type employed in this study was a conversation-oriented (as opposed to assistant type), text-based (as opposed to oral-based) chatbot powered by LLMs. We developed three distinct chatbots using Poe AI (Quora, 2024), a platform that allows users to build customised chatbots without programming knowledge, offering access to a wide range of LLMs (e.g., ChatGPT, Claude). The three chatbots were built using GPT-4o-Mini, with all prompts entered on Poe written in natural language (i.e., English). Several rounds of piloting, which involved prompt revisions and testing, were conducted to ensure that the three chatbots functioned according to the researchers' design and intentions, using the same level of English that was comprehensible to B2-level participants (CEFR) in terms of vocabulary and syntactic structures.

Each chatbot was designed to provide a unique and interactive experience, with differing personalities and feedback styles. Chatbot A, characterised as *"Engaging+ with Basic Feedback,"* offers an emotionally engaging experience and occasionally delivers linguistic

feedback when learners make errors, focusing on the conversation topic of *"Insomnia."* Chatbot B, described as *"Engaging+ with Enhanced Feedback,"* also provides an emotionally engaging experience but delivers linguistic feedback on user utterances at every turn, with the conversation centred on *"Diet."* Chatbot C, labelled as *"Engaging– with Basic Feedback,"* is intentionally distant and impersonal, simulating a less engaging interaction. It provides occasional linguistic feedback when learners make errors and focuses on the topic of *"Phone Addiction."* Figure 3 presents a sample conversation for each chatbot.

Procedure

Participants were assigned to one of six groups and completed a survey assessing their perceived competence, foreign language learning anxiety, motivated behaviours, and WTC. The chatbot-assisted language learning intervention spanned three weeks, with each group completing one conversation task per week using one of three different chatbots. Throughout these interactions, participants engaged in discussions on different topics with their assigned chatbot, with each task lasting approximately five minutes. On average, participants contributed around eight turns per conversation. After completing each task, participants were required to fill out a post-task survey, which measured task-related emotions such as enjoyment and anxiety.



Chatbot A
('Engaging+ with Basic Feedback')

Chatbot B
('Engaging+ with Enhanced Feedback')

Chatbot C
('Engaging– with Basic Feedback')

Figure 3. Excerpts from the conversations with each chatbot.
Note. In this study, "basic feedback" refers to occasional brief reformulations, whereas "enhanced feedback" refers to consistent, turn-level metalinguistic feedback alongside corrected models.

Data analysis

To address the first research question—how learners' WTC and its key factors (perceived competence, anxiety, and motivation) influence task emotions in chatbot-assisted language learning across different chatbot features—we employed structural equation modelling (SEM) using Stata version 16 (StataCorp, 2019) as the primary statistical tool. The model illustrated

in Figure 2 examines the relationships between WTC and its key factors, with the primary dependent variable being task emotion, which is represented as a latent variable comprising task enjoyment and task anxiety. The model also incorporates a categorical variable representing three different chatbot types as moderators influencing the primary dependent variable. Model fit was evaluated using five widely accepted indices: the Chi-square statistic (χ^2), Comparative Fit Index (CFI), Tucker-Lewis Index (TLI), Root Mean Square Error of Approximation (RMSEA), and Standardised Root Mean Square Residual (SRMR). The criteria for an acceptable fit were CFI and TLI values exceeding 0.95, along with RMSEA and SRMR values below 0.05 (Kline, 2016; Lee H. et al., 2020; Lee & Lee, 2023). To address the second research question, we examined task emotion by separating it into two components—task enjoyment and task anxiety—and conducted path analysis. This approach was selected because no latent variables were involved in the model, as depicted in Figure 4. Each component was analysed separately; the top panel of Figure 4 represents the model for task enjoyment, while the bottom panel presents the model for task anxiety. As in the SEM analysis, the model fit was assessed using the same indices and criteria.

Results

Descriptive statistics and correlations

Table 1 presents the descriptive statistics for the primary target variables. Perceived competence, anxiety, motivation, and WTC were measured before the intervention with means of 6.65 (*SD* = 1.34), 3.20 (*SD* = 0.62), 4.58 (*SD* = 1.16), and 4.61 (*SD* = 1.11), respectively. Task enjoyment (*M* = 4.07, *SD* = 1.29) and task anxiety (*M* = 1.75, *SD* = 0.80) were assessed during the intervention. The reported means and standard deviations highlighted the variability within each variable, reflecting individual differences among learners before and during the chatbot-assisted language learning process. Direct comparisons between variables are not valid due to differences in their measurement scales.

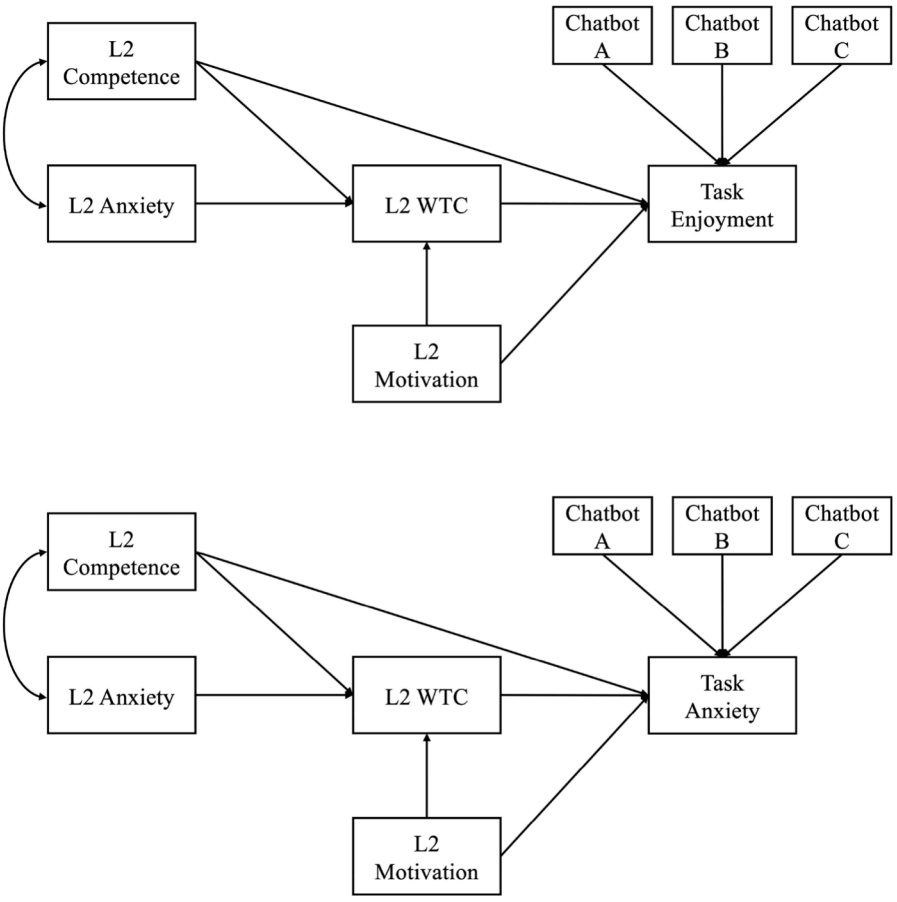


Figure 4. Proposed models of WTC and its high evidence factors in shaping task enjoyment (Top) and task anxiety (Bottom) in chatbot-assisted language learning, incorporating chatbot types.

Table 1. Descriptive statistics for target variables.

	Mean	**SD**	**Minimum**	**Maximum**
Perceived Competence	6.65	1.34	2.8	9
Anxiety	3.20	0.62	1.88	4.5
Motivation	4.58	1.16	2	6
Willingness to Communicate	4.61	1.11	2.3	6
Task Enjoyment	4.07	1.29	1	6
Task Anxiety	1.75	0.80	1	5.5

Table 2 presents the correlations between the target variables, highlighting significant relationships that substantiate their inclusion in the SEM and path analysis while also investigating potential multicollinearity issues. Perceived competence demonstrated strong positive correlations with WTC (r = .69, p < .001) and motivation (r = .49, p < .001) along with a significant negative correlation with anxiety (r = −.37, p < .01). These findings indicate that perceived competence is closely related to both WTC and emotional states. Anxiety was negatively correlated with motivation (r = −.49, p < .001) and WTC (r = −.50, p < .001), indicating its potential role as a suppressor of positive learning-related factors. Task enjoyment was positively correlated with WTC (r = .30, p < .05), indicating that learners with greater

WTC are likely to experience more enjoyment. However, its relatively weak correlations with other variables imply that additional mediators or moderators, such as chatbot features, may influence this relationship. Task anxiety displayed weaker relationships with the other variables, with only a modest negative correlation with perceived competence ($r = -.26$, $p < .05$).

Table 2. Correlations between target variables.

	(1)	(2)	(3)	(4)	(5)	(6)
(1) Perceived Competence	–					
(2) Anxiety	−.37**	–				
(3) Motivation	.49***	−.49***	–			
(4) Willingness to Communicate	.69***	−.50***	.58***	–		
(5) Task Enjoyment	.19	−.22	.18	.30*	–	
(6) Task Anxiety	−.26	.04	−.09	−.13	−.24	–

*** $p < .001$, ** $p < .01$, * $p < .05$.

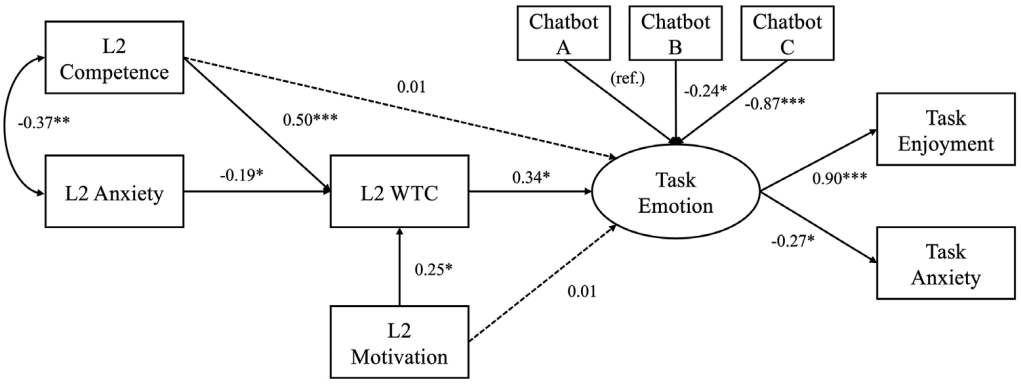


Figure 5. Results of SEM model with task emotion as the latent variable.

Importantly, while the observed correlations support the inclusion of these variables in the SEM and path analyses, the strong correlation between perceived competence and WTC ($r = .69$) warrants caution. High correlations raise concerns about multicollinearity, which may complicate the estimation of unique effects. To address this, multicollinearity diagnostics (i.e., variance inflation factors [VIF] < 10 and 1/VIF > 0.1) were conducted. The findings indicated that the VIF between these two variables was 1.36 and 1/VIF was 0.74, confirming that the variables can be modelled without compromising interpretability. Overall, these relationships align with theoretical expectations and support further exploration of chatbot-assisted language learning.

Structural equation modelling for RQ1

The SEM model (Figure 5) illustrates the contributions of learners' WTC and its key determinants—perceived competence, anxiety, and motivation—on task emotions in chatbot-assisted language learning across various chatbot features. Based on the model fit indices, the model demonstrated a good fit to the data: $\chi^2(8) = 5.78$, $p = 0.67$, RMSEA < .001, CFI > .99, TLI > .99, and SRMR = 0.04.

The results indicate that the relationships between WTC and its key factors align with the findings of the recent meta-analytic structural equation modelling (MASEM) study by Jin and Lee (2022). Specifically, learners' perceived competence and anxiety, which were negatively correlated ($r = -.37$, $p < .01$), jointly influenced WTC ($\beta = 0.50$, $p < .001$; $\beta = -0.19$, $p < .05$, respectively), alongside motivation ($\beta = 0.25$, $p < .05$). However, unlike the study by Jin and Lee (2022), which reported that WTC, perceived competence, and motivation significantly predicted language use (the ultimate dependent variable in general language learning contexts), this study revealed a distinct pattern. Regarding task emotion, the ultimate dependent variable in this study, neither perceived competence ($\beta = 0.01$, $p > .05$) nor motivation ($\beta = 0.01$, $p > .05$) had a significant effect. Instead, only WTC demonstrated a significant contribution to task emotion ($\beta = 0.34$, $p < .05$).

Furthermore, chatbot type significantly moderated task emotion, with learners exhibiting the highest task emotion when engaging with Chatbot A (*"Engaging+ with Basic Feedback"*), followed by Chatbot B (*"Engaging+ with Enhanced Feedback"*) and Chatbot C (*"Engaging– with Basic Feedback"*). The differences between these chatbots were statistically significant (Chatbot A vs. Chatbot B: $\Delta = -0.24$, $p < .05$; Chatbot A vs. Chatbot C: $\Delta = -0.87$, $p < .001$; Chatbot B vs. Chatbot C: $\Delta = -0.63$, $p < .001$).

Finally, when examining the factor loadings of task enjoyment and task anxiety for the latent variable task emotion, task enjoyment ($\beta = 0.90$, $p < .001$) played a significantly larger role compared to task anxiety ($\beta = -0.27$, $p < .05$). These findings suggest that the model is predominantly driven by task enjoyment. A subset model focusing on task anxiety may offer different insights, as explored in the following section.

Path analysis for RQ2a

The first path analysis model (Figure 6) examines how WTC and its key factors influence task enjoyment in chatbot-assisted language learning, considering different chatbot features. The model demonstrated a strong fit to the data, as indicated by the fit indices: $\chi^2(3) = 1.08$, $p = 0.78$, RMSEA < .001, CFI > .99, TLI > .99, and SRMR = 0.01.

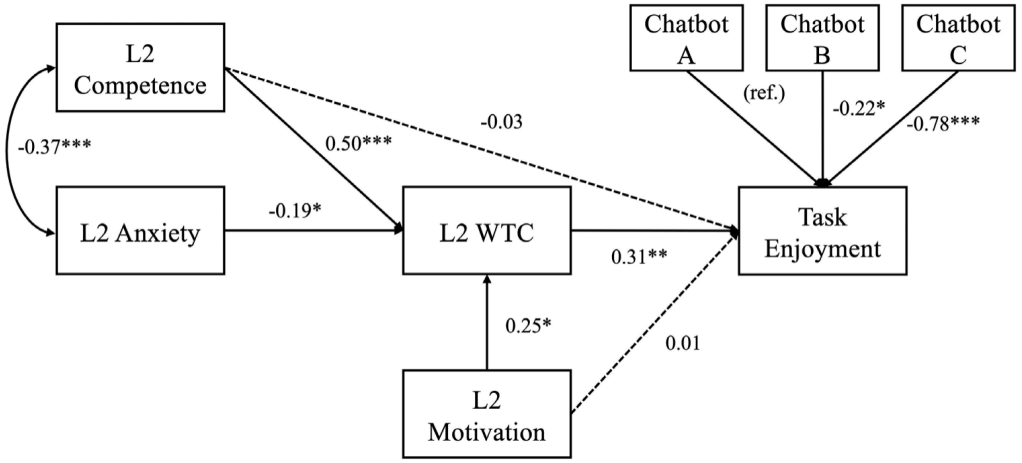


Figure 6. Results of the first path analysis model with task enjoyment as the dependent variable.
*** $p < .001$, ** $p < .01$, * $p < .05$
Note. Path coefficients are standardised.

Consistent with the earlier conjecture, these findings closely mirrored those of the SEM model. Specifically, learners' perceived competence and anxiety, which were negatively correlated ($r = -.37$, $p < .001$), jointly influenced WTC ($\beta = 0.50$, $p < .001$; $\beta = -0.19$, $p < .05$, respectively), alongside motivation ($\beta = 0.25$, $p < .05$). In terms of task enjoyment, neither perceived competence ($\beta = -0.03$, $p > .05$) nor motivation ($\beta = 0.01$, $p > .05$) demonstrated a significant effect, whereas WTC significantly contributed to task enjoyment ($\beta = 0.31$, $p < .01$). Moreover, the chatbot type emerged as a significant moderator of task enjoyment. Learners reported the highest levels of task enjoyment with Chatbot A, followed by Chatbot B and Chatbot C, with statistically significant differences between the chatbots (Chatbot A vs. Chatbot B: $\Delta = -0.22$, $p < .05$; Chatbot A vs. Chatbot C: $\Delta = -0.78$, $p < .001$; Chatbot B vs. Chatbot C: $\Delta = -0.56$, $p < .001$).

For a more precise estimation of the relationships among the variables in the MASEM model, we calculated the direct, indirect, and total effects of each independent variable on WTC and task enjoyment, as outlined in Table 3. The results indicated that competence had a significant indirect effect on task enjoyment ($\beta = 0.16$, $p < .05$), mediated by WTC, while its total effect remained nonsignificant ($\beta = 0.12$, $p > .05$). Regarding anxiety and motivation, we observed that their indirect effects were nonsignificant ($\beta = -0.06$, $p > .05$; $\beta = 0.08$, $p > .05$; respectively). This analysis was particularly necessary because the model figure could not fully capture the indirect effects. For instance, competence, anxiety, and motivation were indirectly related to task enjoyment through WTC, which served as a mediator. Consequently, even though competence and motivation did not demonstrate significant direct effects on task enjoyment, they may still have influenced it indirectly through WTC. Similarly, while anxiety was not directly linked to task enjoyment, its suppressive effect on WTC may have reduced learners' enjoyment of the chatbot-assisted language learning activities.

Path analysis for RQ2b

The second path analysis model (Figure 7) closely resembled the first path analysis, with task anxiety as the ultimate dependent variable. The model exhibited a strong fit to the data, as indicated by the fit indices: $\chi^2(3) = 0.78$, $p = 0.99$, RMSEA < .001, CFI > .99, TLI > .99, and SRMR = 0.01.

Table 3. Total effect of independent variables on WTC and task enjoyment.

Dependent Variable	Independent Variable		Direct Effect	Indirect Effect	Total Effect
Willingness to Communicate	Competence		.50***	NA	.50***
	Anxiety		−.19*	NA	−.19
	Motivation		.25	NA	.25
Task Enjoyment	Willingness to Communicate		.31**	NA	.31
	Competence		−.03	.16	.12
	Anxiety		NA	–.06	−.06
	Motivation		.01	.08	.09
	Chatbot Type	A	(ref)	NA	(ref)
		B	−.22	NA	−.22
		C	−.78***	NA	−.78***

*** $p < .001$, ** $p < .01$, * $p < .05$

Note. Values are standardised coefficients.

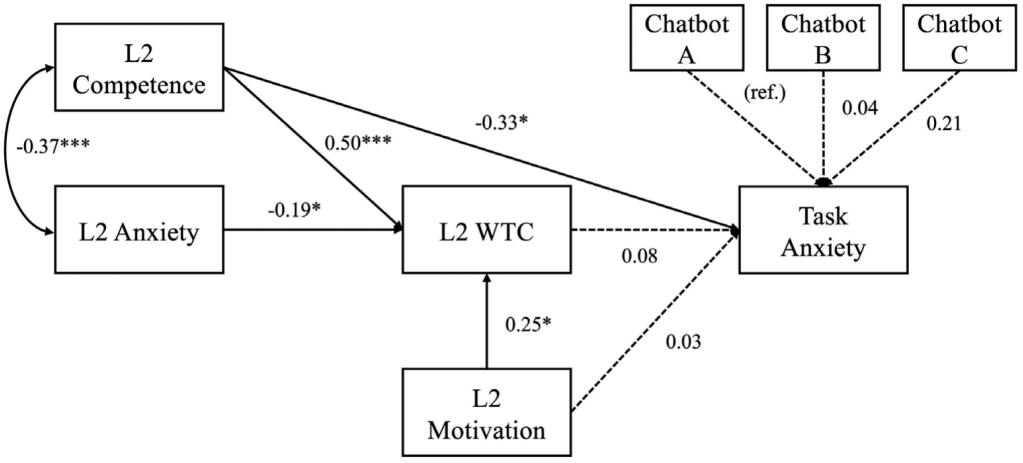


Figure 7. Results of the second path analysis model with task anxiety as the dependent variable.
*** $p < .001$, ** $p < .01$, * $p < .05$

Note. Path coefficients are standardised.

Table 4. Total effect of independent variables on WTC and task anxiety.

Dependent Variable	Independent Variable		Direct Effect	Indirect Effect	Total Effect
Willingness to Communicate	Competence		.50***	NA	.50***
	Anxiety		−.19*	NA	−.19
	Motivation		.25	NA	.25
Task Anxiety	Willingness to Communicate		.08	NA	.08
	Competence		−.33	.04	−.29
	Anxiety		NA	−.02	−.02
	Motivation		.03	.02	.05
	Chatbot Type	A	(ref)	NA	(ref)
		B	.04	NA	.04
		C	.21	NA	.21

*** $p < .001$, ** $p < .01$, * $p < .05$

Note. Values are standardised coefficients.

Except for the relationships between WTC and its key factors, such that learners' perceived competence and anxiety, which were negatively correlated ($r = −.37$, $p < .001$), jointly influenced WTC ($\beta = 0.50$, $p < .001$; $\beta = −0.19$, $p < .05$, respectively), alongside motivation ($\beta = 0.25$, $p < .05$), the second path analysis model revealed substantially different findings. First, with regard to task anxiety, neither WTC ($\beta = 0.08$, $p > .05$) nor motivation ($\beta = 0.03$, $p > .05$) showed a significant effect, whereas perceived competence significantly contributed to task anxiety ($\beta = −0.33$, $p < .05$) in a negative direction. Second, regarding chatbot type, the results indicated that learners exhibited similar levels of task anxiety regardless of different chatbots (Chatbot A vs. Chatbot B: $\Delta = 0.04$, $p > .05$; Chatbot A vs. Chatbot C: $\Delta = 0.21$, $p > .05$; Chatbot B vs. Chatbot C: $\Delta = 0.28$, $p > .05$). Overall, this path analysis model indicated that learners' perceived competence was the only significant predictor of task anxiety in chatbot-assisted language learning. This finding was further reinforced by the indirect effects of the variables within the model. As demonstrated in Table 4, which presents the direct, indirect, and total effects of each independent variable on WTC and task anxiety, the findings indicated that competence, anxiety, and motivation did not exhibit significant indirect effects. Instead, learners' competence remained the sole variable with a significant contributions on task anxiety ($\beta = −0.29$, $p < .05$).

Discussion

This study examined the relationship between WTC and its key factors—perceived competence, foreign language anxiety, and motivation—in addition to task emotions in chatbot-assisted language learning, utilising various chatbot features. These findings provide critical insights into the role of WTC in shaping learners' emotional experiences, the influence of chatbot design on task enjoyment, and the influence of perceived competence on task

anxiety. Collectively, these insights contribute to a deeper understanding of how digital language learning tools can either enhance or hinder learner engagement and emotional well-being.

WTC as a key predictor of task enjoyment

A central finding of this study is that WTC significantly influences learners' task enjoyment, reinforcing the idea that those who are more willing to communicate tend to have more positive emotional experiences during chatbot-assisted language learning. This finding aligns with previous research emphasising the role of WTC in facilitating language use and engagement (Jin & Lee, 2022). However, our study extends this perspective by demonstrating that WTC not only predicts language use, but also shapes learners' emotional states during learning tasks. Learners with a higher WTC are more likely to perceive communication-oriented tasks as opportunities rather than challenges, engaging in them with a more positive mindset (Lou & Noels, 2019). Consequently, this enhances their enjoyment of chatbot-assisted language learning. Conversely, those with lower WTC may approach tasks reluctantly, leading to reduced enjoyment and potentially limiting their overall engagement in chatbot interactions.

These findings suggest that fostering WTC may be an effective strategy for improving learner engagement and emotional experiences in AI-driven language learning environments. Considering that enjoyment plays a crucial role in sustaining motivation and participation (Dewaele & MacIntyre, 2014), integrating features that encourage WTC, such as interactive prompts, adaptive feedback, and peer collaboration *via* chatbots, can enhance the overall learning experience. By creating a learning environment in which one feels comfortable and motivated to communicate, chatbot-assisted language learning can be both more effective and enjoyable.

The influence of chatbot features on task enjoyment

Our findings indicate that different chatbot features led to varying levels of task enjoyment, highlighting the significance of engagement and feedback in shaping learner experiences. Among the three chatbot conditions, Chatbot A ("Engaging + with Basic Feedback") yielded the highest enjoyment, followed by Chatbot B ("Engaging + with Enhanced Feedback"), while Chatbot C ("Engaging– with Basic Feedback") resulted in the lowest enjoyment. This pattern indicates that learners preferred an engaging chatbot environment, but were also sensitive to the level of feedback provided. Overall, given that the topics addressed by the three chatbots were largely familiar to the participants and thus unlikely to have influenced their emotional responses, it can be suggested that the combination of Chatbot A's engagement-enhancing features—such as interactive dialogue and immersive environments (Belda-Medina &

Kokošková, 2023)—and its basic level of feedback likely contributed to higher enjoyment compared to the other two chatbots. In particular, the lower level of enjoyment reported for Chatbot B, compared to Chatbot A, may be associated with the notion that excessive feedback can overwhelm learners (e.g., Fyfe et al., 2015), increasing task-related demands and potentially leading to anxiety (also, less enjoyment).

Furthermore, the notably lower enjoyment experienced in the less engaging environment of Chatbot C reinforces the notion that interactivity and engagement are essential for a positive learning experience (e.g., Dewaele & MacIntyre, 2014; Li et al., 2023). Learners may feel more motivated and emotionally invested in dynamic, responsive, or contextually rich chatbot interactions. This highlights the necessity of chatbot designs that strike a balance between engagement and feedback intensity. By optimising both these elements, chatbot-assisted language learning can foster enjoyable and effective learning.

Perceived competence and the vicious cycle of language anxiety

A key finding of this study is that task anxiety was primarily influenced by perceived competence, with learners who perceived themselves as less competent in the target language experiencing higher anxiety levels. This supports the well-established concept of the vicious cycle of language anxiety (Lou & Noels, 2020), wherein learners with low perceived competence feel anxious about using the language, which, in turn, negatively affects their performance, further diminishing their perceived competence and perpetuating the cycle of anxiety and low confidence. In the context of chatbot-assisted learning, this cycle may be particularly pronounced, as learners with lower competence may feel increased pressure to respond accurately to chatbots, thereby exacerbating their anxiety.

Interestingly, task anxiety was not significantly affected by the WTC, indicating that even learners with a strong WTC may still experience anxiety if they perceive their competence as low. This highlights the significance of considering learners' perceived competence when designing chatbot-based learning environments. While chatbot-assisted language learning is often viewed as a low-pressure alternative to traditional classroom settings (e.g., Alrajhi, 2024; Kohnke, 2023b), this finding indicates that learners with low self-perceived competence may still face considerable anxiety in AI-driven environments.

The lack of significant effects of the chatbot type on task anxiety further reveals that engagement features and feedback variations primarily influence learners' enjoyment rather than their anxiety levels. Consequently, reducing task anxiety may require interventions beyond chatbot design, such as implementing adaptive scaffolding strategies, providing confidence-building exercises, and incorporating explicit anxiety-reduction techniques into the learning process (e.g., Lou & Noels, 2016). Chatbots could greatly benefit from features that

support learners' confidence, such as adjustable task difficulty levels, reassurance mechanisms, and personalised feedback, which gradually build learners' self-efficacy (Kuhail et al., 2023). These strategies can help mitigate the negative emotional influence of anxiety and foster a more supportive and effective learning environment for learners with varying levels of perceived competence.

Conclusion

In summary, this study highlights the significance of psychological factors in shaping learners' emotional experiences during chatbot-assisted language learning. Although WTC emerged as a key predictor of task enjoyment, perceived competence was the sole significant predictor of task anxiety. Additionally, chatbot features played a pivotal role in determining enjoyment, with an engaging environment and moderate feedback fostering positive experiences.

Nevertheless, there are some limitations of the study that warrant discussion. Above all, most of them were related to the small sample size. The first one concerns the inability to fully utilise the dataset. The task emotion construct originally encompassed enjoyment, anxiety, boredom, and curiosity. However, given the limited sample of 66 (22 participants across three tasks), creating a latent variable with four observed indicators was not statistically feasible. Consequently, we focused on the more representative emotional components of enjoyment and anxiety, excluding boredom and curiosity, in our structural model. Second, the restricted sample size prevented us from incorporating various important learner-level covariates, such as gender, prior chatbot experience, and personality. These factors could potentially influence WTC or task emotion, as well as their interplay. Third, it should also be mentioned that, given the short task duration and the text-only modality, findings may not generalise to longer or spoken interactions. Finally, differences in conversation topics across chatbots may have had a minor influence on enjoyment, although the design emphasised chatbot persona and feedback type.

For a future direction, we suggest that future research with a larger sample size should explore a more comprehensive range of task emotions and the roles of individual differences, offering a deeper understanding of the emotional dynamics in chatbot-assisted language learning. Also, future research should aim to disentangle the chatbot features explored in this study and investigate the optimal chatbot design for individual students with varying attributes, tailored to their specific needs and characteristics. Longitudinal studies could also track changes in WTC and emotions over time through sustained engagement in chatbot-assisted language learning. By advancing our understanding of the interplay between WTC, emotions, and chatbot features, this study contributes to the ongoing discussion on how AI-assisted tools can

be designed to be emotionally supportive, engaging, and adaptable to the diverse needs of learners by considering human elements (Kohnke & Moorhouse, 2025), in addition to being effective for language learning.

Author contributions: CRediT

Hansol Lee: Conceptualisation; Formal analysis; Methodology; Writing—original draft

Jang Ho Lee: Methodology; Writing—original draft; Writing—reviewing and editing

Disclosure statement

No potential conflict of interest was reported by the author(s).

Funding

This research was supported by the Hwarangdae Research Institute at the Korea Military Academy (Award No. 2025B1021).

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